QuadDrone - calculation report

500-class GPS quadcopter, 3D-printed PA6-CF frame, Inventor 2027 parametric model (QuadDrone.iam).

1. MASS (Inventor MassProperties; purchased parts carry datasheet mass overrides)

CAD mass measured	1616.2 g	CG [0.7, 0.07, 6.33] mm
unmodelled allowance	80.0 g	(wiring/XT60 45, RX 5, telemetry 22, ties 8)
ALL-UP MASS (AUW)	1696.2 g	
Printed frame (PA6-CF)	473.6 g	
Propulsion (motors, props, ESCs)	332.0 g	
Fasteners, standoffs, dampers, foam, straps	109.3 g	
Avionics + payload (FC, GPS, PM, gimbal)	221.3 g	
Battery	480.0 g	

2. THRUST-TO-WEIGHT (T-Motor AIR2216 II KV920 + T1045 bench table, store.tmotor.com goods id=1220)

max static thrust per motor	1332 g @16.0 V, 1140 g @14.8 V (x(14.8/16)^2)
TWR = 4 Tmax / AUW	3.14 (full charge) / 2.69 (nominal)
full-throttle current	65.5 A total = 13.1 C (pack rated 45 C)

3. HOVER ENDURANCE

hover thrust per motor	424 g (48% throttle @16 V, ~5631 rpm)
power per motor (fit)	55.2 W (7.68 g/W)
avionics	8 W (FC+GPS, gimbal hold, RX/telemetry, ESC idle)
total	229 W -> 15.47 A @14.8 V
usable energy	4.0 Ah (80% of 5000 mAh)
HOVER TIME	15.5 min (cross-check with SunnySky X2212 data: 14.6 min)

4. STRUCTURAL FEA - solver: gmsh 4.15 (STEP from Inventor, OpenCASCADE, tetra mesh) + scikit-fem 12 (linea
elasticity, quadratic P2 tetrahedra) + Intel MKL PARDISO direct solve (pypardiso). NOT Inventor Stress A
Material PA6-CF (Bambu Lab): E 3560 MPa (conditioned bending modulus), nu 0.35, rho 1.09 g/cm3.
Allowable 45 MPa = conditioned layer (Z) bending strength (most conservative); UTS XY dry 102 MPa.

Case A arm, full thrust 13.07 N + torque 202 N.mm, root clamped where the plates grip it
mesh 87,140 tets / 501,357 DOF; peak vM 9.65 MPa (rigid-clamp edge singularity),
6.19 MPa away from the clamp, 99.9th pct 4.46 MPa
motor-seat deflection 0.803 mm, tip 0.982 mm
SF 4.7 (peak, 45 MPa) / 7.3 (away from clamp) / 10.6 vs 102 MPa

Case B 3 g manoeuvre (thrust 3*AUW*g/4 = 12.48 N per motor), centre frame: both plates +
4 arm roots cut at r = 110 mm carrying shear 9.35 N + moment 1355 N.mm each,
payload inertia x3 at the FC/hatch and gimbal mounts, frame self-weight x3, reacted at the battery-cage
mesh 147,460 tets / 797,445 DOF; peak vM 5.06 MPa (arm roots); bottom plate 3.67, top plate 2.26 MPa
mode 1: 67.3 Hz lateral (in-plane) deflection 0.125 mm, SF 8.9 vs 45 MPa

Case M arm modes, clamped root, motor+prop+nut+screws as tip mass (80 g):
mode 2: 79.4 Hz lateral (in-plane)
mode 3: 507.6 Hz vertical bending (flap)
mode 4: 588.7 Hz vertical bending (flap)

hover rotor frequency 1P = 94 Hz, blade pass 2P = 188 Hz -> first flap mode sits
below 1P (supercritical, frequency ratio ~1.4): passes through resonance only during spool-up; set a PX
notch filter near the flap mode if logs show it. Raising ArmH 20->26 mm lifts f1 ~45% for ~+15% arm mas

Mesh check (case A): HXT mesh 492k DOF -> 9.98 / 6.18 MPa, 0.804 mm; Delaunay 501k DOF -> 9.65 / 6.19 MPa
0.803 mm: nominal stress and deflection converged (<1%), only the singular clamp-edge peak moves.

5. ASSEMBLY CHECKS (Inventor API)

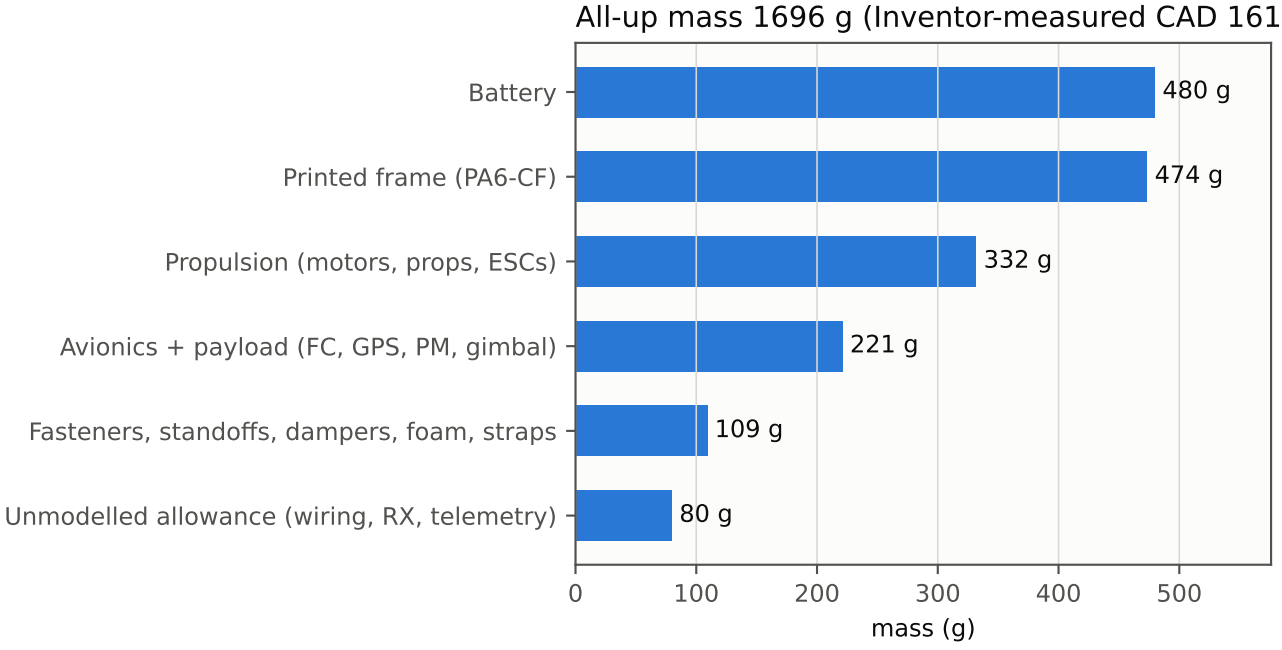
139 occurrences / 31 files, 221 constraints {'Flush': 111, 'Mate': 12, 'Insert': 98}
DOF {'0T/0R': 139}, unhealthy constraints 0, interference 0 pairs (0.000 mm3)

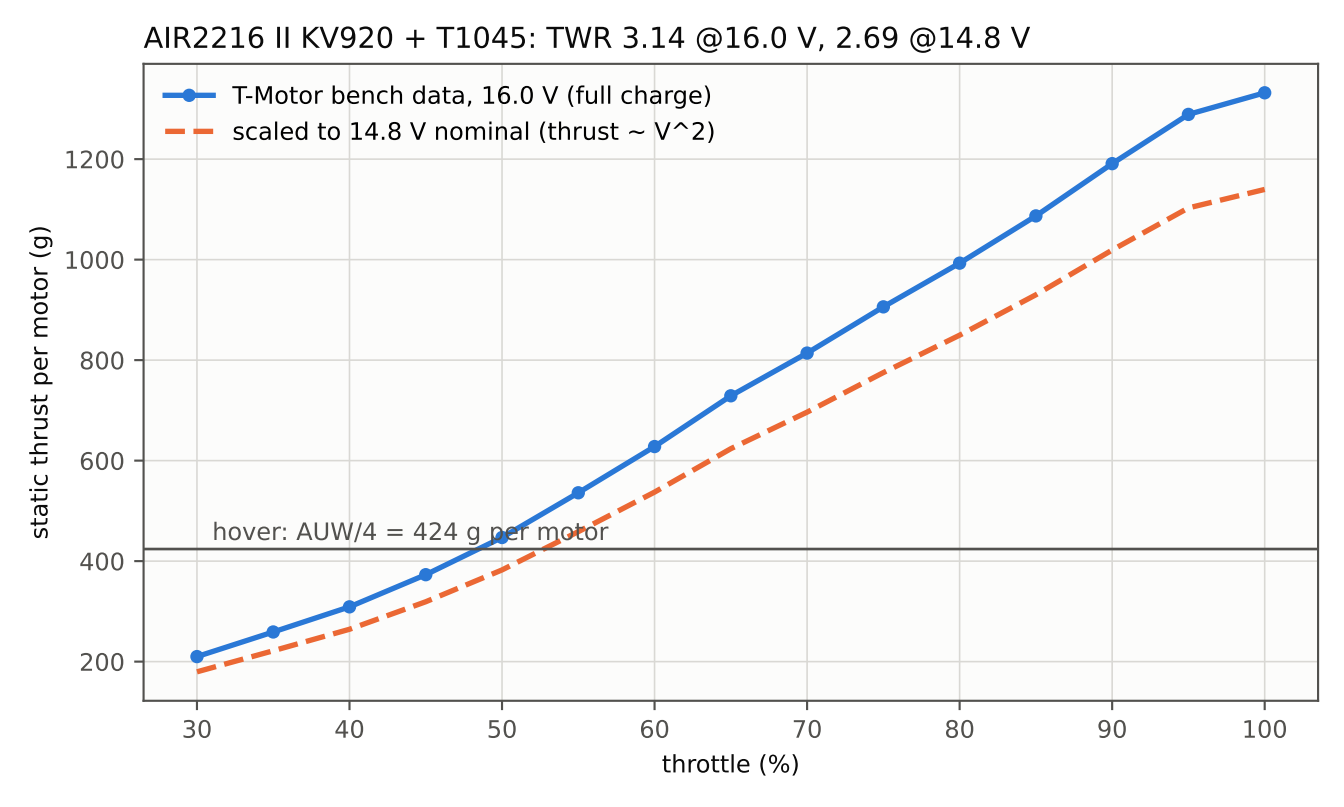
6. COMPONENT CAD SOURCES

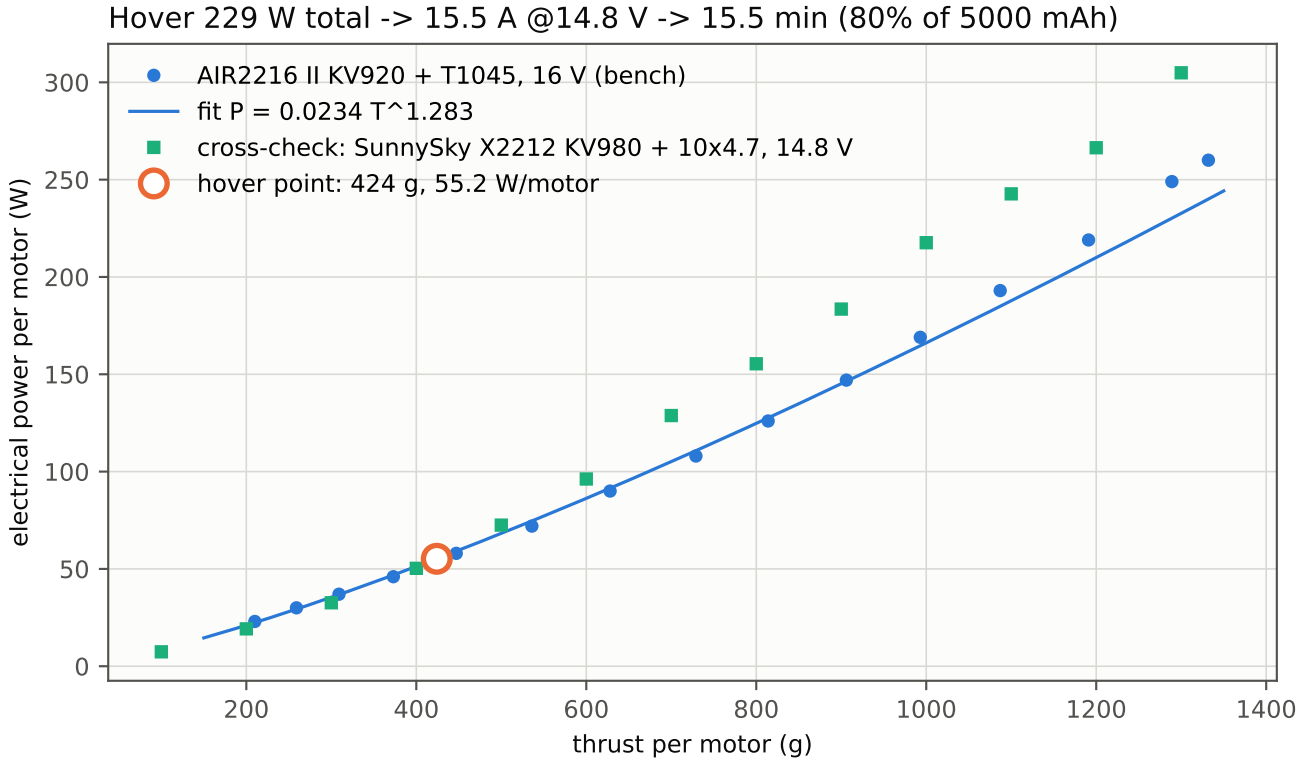
vendor STEP (downloaded): Holybro Pixhawk 6C, Holybro M9N GPS, T-Motor AIR2216 II KV920 motor
modelled to datasheet: T1045 props (twisted loft), 30 A ESC (Hobbywing XRotor Micro size), Gens Ace 4S 50
battery, PM02-class power module, 2-axis micro gimbal + camera (mass estimated 110 g), dampers, strap, fa

7. ASSUMPTIONS / LIMITS

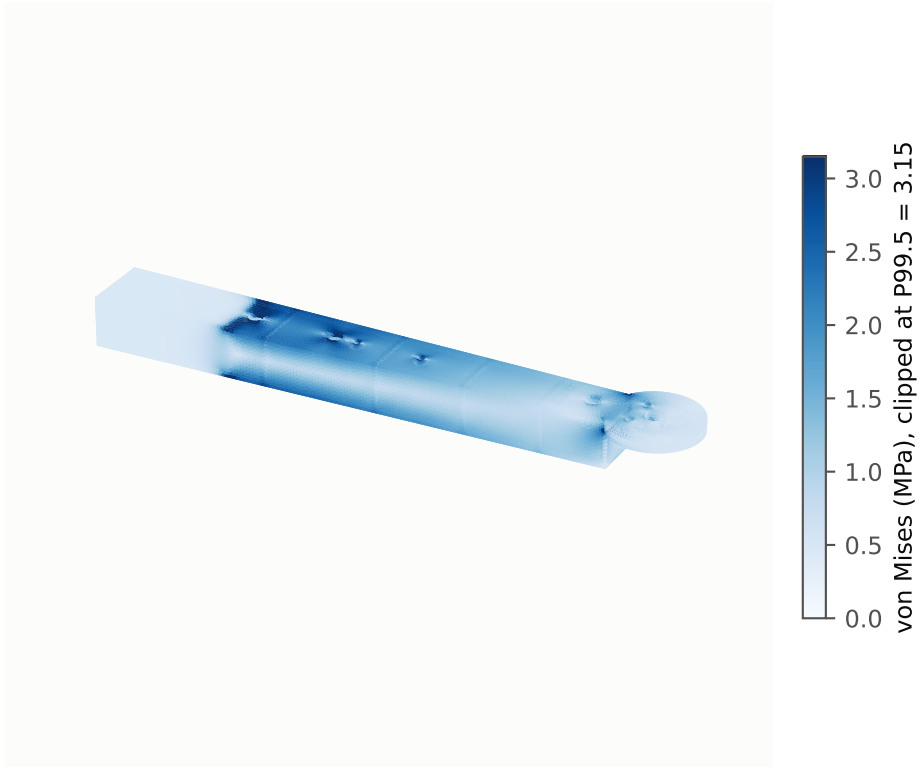
- printed parts modelled solid; real prints with 40% infill in thick zones will be lighter and less stiff
- linear elastic, isotropic PA6-CF; layer direction treated via the 45 MPa conditioned Z allowable
- case B: plates and arm roots bonded (clamped bolts idealised as plugged holes); arms beyond r=110 mm re
  by their statically exact shear + moment; battery reaction applied at its 4 standoff holes
- endurance: static bench data, hover only, no wind/forward flight; 14.8 V average; 80% depth of discharg
- original plan listed a 2212 920KV motor; the real T-Motor AIR2216 II KV920 STEP was used instead (4S/10
  a 2212 on 4S + 10 in exceeds its continuous current at full throttle). SunnySky X2212 data shown as cro


All-up mass 1696 g (Inventor-measured CAD 1616
Battery
480 g
Printed frame (PA6-CF)
474 g
Propulsion (motors, props, ESCs)
332 g
Avionics + payload (FC, GPS, PM, gimbal)
221 g
Fasteners, standoffs, dampers, foam, straps
109 g
Unmodelled allowance (wiring, RX, telemetry)
80 g
0
100
200
300
400
500
mass (g)


AIR2216 II KV920 + T1045: TWR 3.14 @16.0 V, 2.69 @14.8 V
T-Motor bench data, 16.0 V (full charge)
scaled to 14.8 V nominal (thrust ~ V^2)
hover: AUW/4 = 424 g per motor
static thrust per motor (g)
throttle (%)
200
400
600
800
1000
1200
30
40
50
60
70
80
90
100

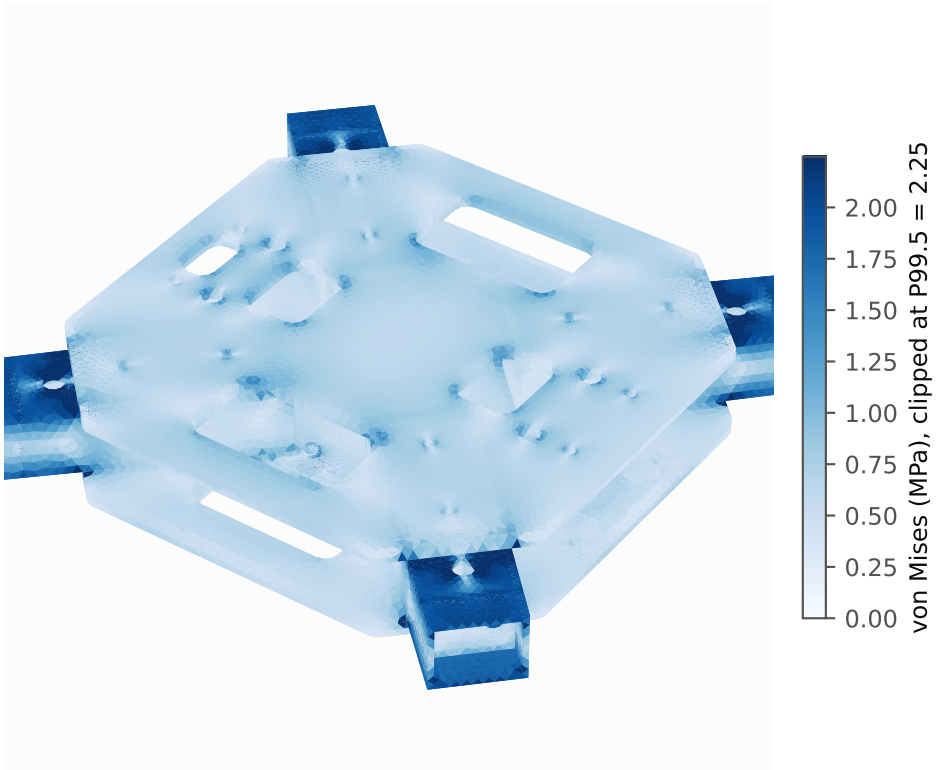

Hover 229 W total -> 15.5 A @14.8 V -> 15.5 min (80% of 5000 mAh)
AIR2216 II KV920 + T1045, 16 V (bench)
fit P = 0.0234 T^1.283
cross-check: SunnySky X2212 KV980 + 10x4.7, 14.8 V
hover point: 424 g, 55.2 W/motor
electrical power per motor (W)
thrust per motor (g)
0
50
100
150
200
250
300
200
400
600
800
1000
1200
1400

Case A - arm at full thrust 13.07 N + 202 N.mm, root clamped
peak 9.65 MPa (clamp edge), 6.19 MPa elsewhere



FEA: gmsh + scikit-fem (P2 tetra) + PARDISO. Colour = von Mises, sequential scale clipped at the 99.5th percentile.

Case B - 3 g manoeuvre, centre frame + arm roots
peak 5.06 MPa, max deflection 0.125 mm



FEA: gmsh + scikit-fem (P2 tetra) + PARDISO. Colour = von Mises, sequential scale clipped at the 99.5th percentile.